

YZV202E Optimization for Data Science Project Proposal

Optimizing Fire Station Expansion in Istanbul to Reduce Emergency Response Time

Hikmet Gültekin
AI and Data Engineering
Istanbul Technical University
gultekinh22@itu.edu.tr
150220321

Ata Türkoğlu
AI and Data Engineering
Istanbul Technical University
turkoglua21@itu.edu.tr
150210337

Azizagha Majidov
AI and Data Engineering
Istanbul Technical University
majidov23@itu.edu.tr
150230910

I. PROJECT DESCRIPTION

Istanbul is a dense megacity with high seismic risk, heavy traffic, and strong spatial variation in emergency demand. In fire and rescue operations, a few minutes of delay can materially change the outcome, so the placement of stations is a public-safety optimization problem rather than only an infrastructure decision. This project studies whether the current fire station network is aligned with district-level demand and identifies where a limited number of additional stations should be opened to reduce expected response time. Using open data from the Istanbul Metropolitan Municipality (IBB), we will build a reproducible location-allocation pipeline and maintain the implementation in a GitHub repository: <https://github.com/hikmedit/optimization-project>.

II. PROBLEM DEFINITION

We formulate the expansion problem as a weighted p -median facility-location model with fixed existing stations, following the classical location-allocation literature [1]–[3]. We use the p -median objective as the primary model because it improves the full distribution of response times instead of optimizing only the zones that fall inside a chosen coverage threshold.

Let I be the set of demand zones and J the set of candidate station sites. Each demand zone $i \in I$ has weight w_i , derived from district population and stress-tested with historical fire incident frequency. Let t_{ij} denote the estimated travel time from demand zone i to site j . Parameter $e_j \in \{0, 1\}$ indicates whether site j already contains an active station. Decision variable $x_j \in \{0, 1\}$ is 1 if site j is active after optimization, and $y_{ij} \in \{0, 1\}$ is 1 if demand zone i is assigned to site j .

If exactly p new stations can be opened, the model is

$$\begin{aligned} \min \quad & \sum_{i \in I} \sum_{j \in J} w_i t_{ij} y_{ij} \\ \text{s.t.} \quad & \sum_{j \in J} y_{ij} = 1, & \forall i \in I \\ & y_{ij} \leq x_j, & \forall i \in I, \forall j \in J \\ & x_j \geq e_j, & \forall j \in J \\ & \sum_{j \in J} (x_j - e_j) = p \\ & x_j \in \{0, 1\}, y_{ij} \in \{0, 1\}. \end{aligned} \tag{1}$$

The target variable is the weighted average emergency response time implied by (1). As a secondary performance measure, we will report the share of weighted demand served within a policy threshold such as $T_{\max} = 5$ or 8 minutes, which connects the model to maximal covering ideas used for emergency facilities [2], [3].

To bridge the gap between discrete allocation and continuous optimization, we extend the formulation into a continuous space. For a candidate district selected by the discrete model, the exact coordinates (x, y) of the new station will be treated as continuous design variables. This allows us to formulate a continuous objective function based on geometric distance metrics, enabling the mathematical analysis of its convexity and the application of gradient-based optimization methods to find the optimal local placement.

III. DATASET

The main data source will be the IBB Open Data Portal [5]. We plan to combine four groups of datasets: existing fire station coordinates, district population statistics, fire incident counts and average arrival-time records. To refine travel-time estimates, we will also use the first-degree emergency transportation routes published by IBB. These sources are appropriate because they directly capture both the spatial

supply side (station locations and network access) and the demand side (population and incident intensity).

In the first project iteration, districts will be represented by centroid-based demand zones to keep the optimization tractable and interpretable. Candidate sites will consist of current station locations and district centroids that have reasonable access to emergency routes. After cleaning coordinate systems and matching district names across files, we will build a modeling table containing the demand weights w_i , the candidate-site list, and the travel-time matrix t_{ij} . Distance will be computed either on the road network or through a calibrated geometric proxy, and the calibration will be checked against reported average arrival times. If time permits, the same pipeline can later be refined from district centroids to a finer grid without changing the core optimization model.

IV. METHODOLOGY

Our workflow follows a hybrid discrete-continuous optimization pipeline consisting of four stages. First, we will preprocess the GIS layers, estimate district-to-site travel times, and create a baseline map of current accessibility. Second, we will solve the mixed-integer linear programming model in (1) with PuLP or OR-Tools to obtain an exact benchmark for small and medium candidate sets. Third, we will implement two metaheuristics, Genetic Algorithm (GA) and Simulated Annealing (SA), because the search space grows combinatorially as the number of candidate sites increases [4].

Fourth, we will integrate continuous optimization techniques to fine-tune the final station coordinates. Once the discrete models (MILP/GA) identify the best candidate regions, we will use gradient-based methods, specifically Steepest Descent and Newton's Descent, to minimize the travel time within that continuous space. During these descent updates, 1D search methods such as Golden Section or Bisection will be implemented to optimize the step size. We will also analyze the Hessian matrix of the distance function to verify its convexity and confirm whether the optimized coordinates represent a global minimum. Additionally, a Bisection search can be applied as an outer loop to find the minimum required budget p for a specific coverage threshold.

We will compare MILP, GA, and SA by objective value, runtime, and service coverage for several budget levels such as $p \in \{1, 3, 5\}$. The notebook will include maps of current and proposed stations, assignment visualizations, and interactive controls for p , the demand-weight definition, and an optional road-disruption multiplier for earthquake scenarios. This design keeps the proposal focused on one primary optimization target while leaving room for richer scenario analysis.

REFERENCES

- [1] S. L. Hakimi, "Optimum locations of switching centers and the absolute centers and medians of a graph," *Operations Research*, vol. 12, no. 3, pp. 450–459, 1964.
- [2] C. Toregas, R. Swain, C. ReVelle, and L. Bergman, "The location of emergency service facilities," *Operations Research*, vol. 19, no. 6, pp. 1363–1373, 1971.
- [3] R. Church and C. ReVelle, "The maximal covering location problem," *Papers in Regional Science*, vol. 32, no. 1, pp. 101–118, 1974.
- [4] S. Bolouri, A. Vafaeinejad, A. A. Alesheikh, and H. Aghamohammadi, "The ordered capacitated multi-objective location-allocation problem for fire stations using spatial optimization," *ISPRS International Journal of Geo-Information*, vol. 7, no. 2, p. 44, 2018.
- [5] Istanbul Metropolitan Municipality, "IBB Open Data Portal." [Online]. Available: <https://data.ibb.gov.tr/>. Accessed: Apr. 17, 2026.